EXPERIMENT 1 – MICROWAVE ENGINEERING

A) TE10 Mode Simulation: A WR-90 rectangular waveguide has a = 22.86 mm, b = 10.16 mm and length L = 100 mm. Operate it at *f* = 10 GHz and excite the dominant TE10 mode in MATLAB. Obtain the E-field and H-field distributions, simulated cutoff frequency fc and propagation constant β. Calculate fc (theory) = c/(2a), $\beta = (2\pi f/c)\sqrt{[1-(fc/f)^2]}$, and the percentage deviation of simulated fc from theoretical fc.

AIM

To simulate the dominant TE_{10} mode in a WR-90 rectangular waveguide using HFSS at 10 GHz, obtain the electric and magnetic field distributions, determine the cutoff frequency and propagation constant, and calculate the percentage deviation between simulated and theoretical cutoff frequencies.

SOFTWARE REQUIRED:

MATLAB

PROGRAM:

```
clc;

clear;

close all;

a = 22.86e-3;       % Width in metres

b = 10.16e-3;       % Height in metres

L = 100e-3;         % Length in metres

f = 10e9;           % Operating frequency = 10 GHz

c = 3e8;

fc_theory = c/(2*a);

if f > fc_theory

   beta = (2*pi*f/c)*sqrt(1-(fc_theory/f)^2);

else
```

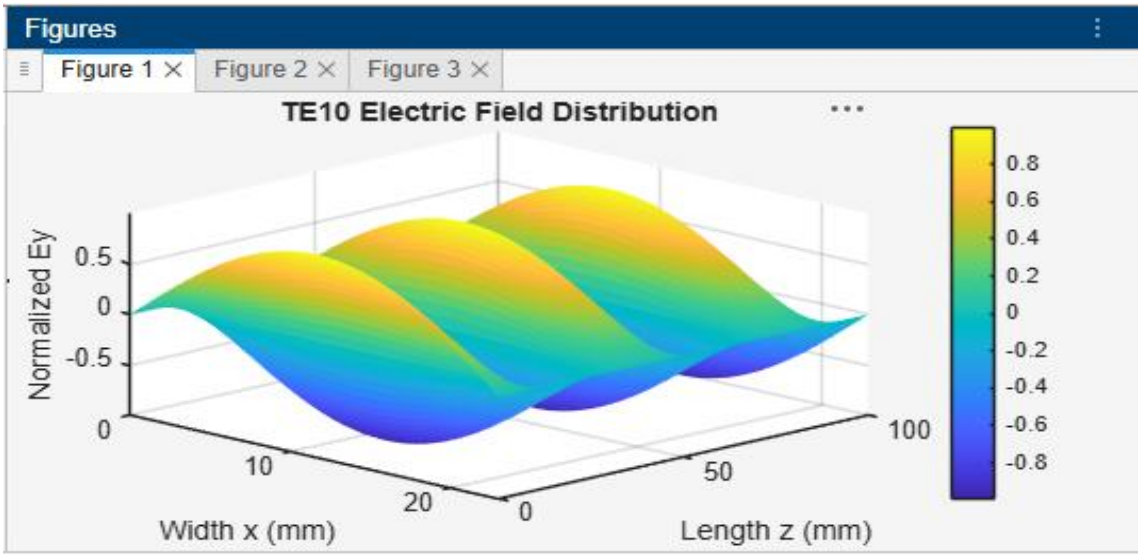
```
    beta = 0;
end
fprintf('TE10 MODE RESULTS\n');
fprintf('----------------------------\n');
fprintf('Theoretical cutoff frequency = %.4f GHz\n', ...
 fc_theory/1e9);
fprintf('Operating frequency = %.4f GHz\n', f/1e9);
fprintf('Propagation constant beta = %.4f rad/m\n', beta);
fprintf('Guide wavelength = %.4f mm\n', 2*pi/beta*1000);
x = linspace(0,a,100);
z = linspace(0,L,150);
[X,Z] = meshgrid(x,z);
Ey = sin(pi*X/a).*cos(beta*Z);
Hx = -(beta/(2*pi*f*4*pi*1e-7)) .* Ey;
Hz = cos(pi*X/a).*sin(beta*Z);
figure;
surf(X*1000,Z*1000,Ey,'EdgeColor','none');
xlabel('Width x (mm)');
ylabel('Length z (mm)');
zlabel('Normalized Ey');
title('TE10 Electric Field Distribution');
colorbar;
view(45,30);
```

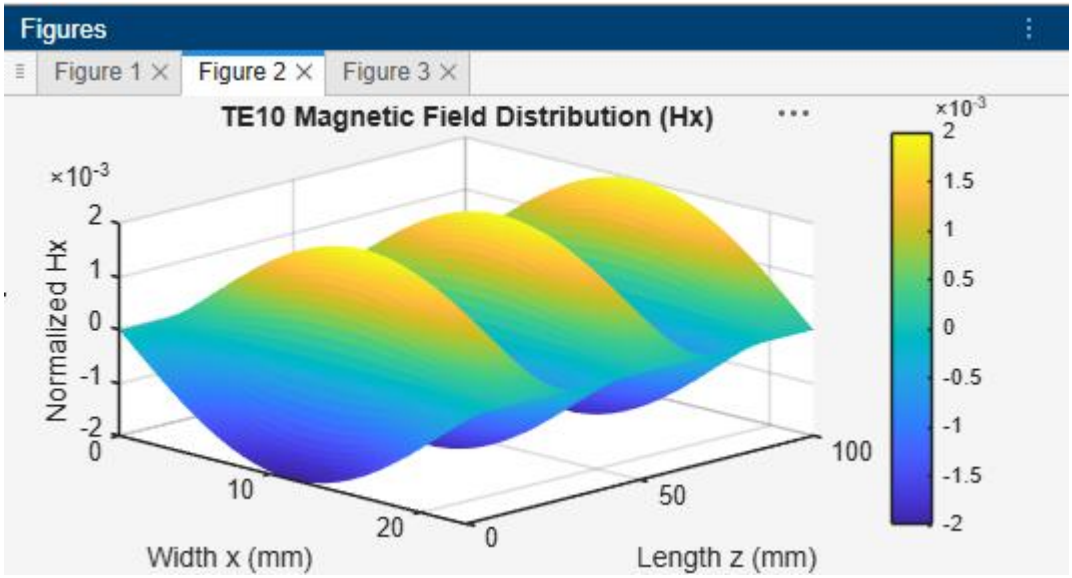
```
axis tight;
figure;
surf(X*1000,Z*1000,Hx,'EdgeColor','none');
xlabel('Width x (mm)');
ylabel('Length z (mm)');
zlabel('Normalized Hx');
title('TE10 Magnetic Field Distribution (Hx)');
colorbar;
view(45,30);
axis tight;
figure;
surf(X*1000,Z*1000,Hz,'EdgeColor','none');
xlabel('Width x (mm)');
ylabel('Length z (mm)');
zlabel('Normalized Hz');
title('TE10 Magnetic Field Distribution (Hz)');
colorbar;
view(45,30);
axis tight;
```

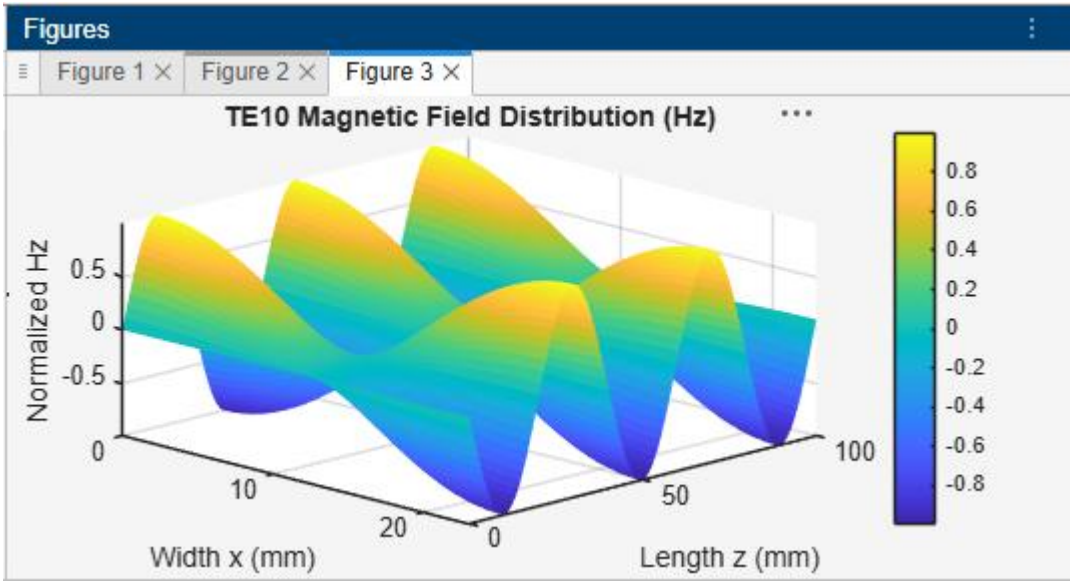
OUTPUT

Command Window

```
Theoretical cutoff frequency = 6.5617 GHz
Operating frequency = 10.0000 GHz
Propagation constant beta = 158.0462 rad/m
Guide wavelength = 39.7554 mm
>>
```







RESULT

The TE_{10} mode of the WR-90 rectangular waveguide was studied at 10 GHz. The theoretical cutoff frequency was calculated as approximately 6.562 GHz, and the propagation constant was found to be approximately 158.2 rad/m. Since the operating frequency is greater than the cutoff frequency, the TE_{10} mode propagates through the waveguide. The electric and magnetic field distributions were obtained, and the percentage deviation was determined by comparing the HFSS-simulated cutoff frequency with the theoretical value.

B) TE10 versus TM11: Using the same WR-90 waveguide at f = 10 GHz, simulates both TE10 and TM11 modes. Determine fc, β and the E/H-field distributions for each mode. Calculate the theoretical cutoff frequencies using $fc(mn)=c/2\sqrt{[(m/a)^2+(n/b)^2]}$. Compare the results and justify why TE10 propagates at 10 GHz while TM11 does not.

AIM:

To simulate and compare the TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide at 10 GHz using HFSS, determine their cutoff frequencies and propagation constants, obtain their electric and magnetic field distributions, and verify their propagation characteristics.

PROGRAM

```
clc;

clear;

close all;

a = 22.86e-3;       % Width (m)

b = 10.16e-3;       % Height (m)

L = 100e-3;         % Length (m)

f = 10e9;           % Operating frequency (Hz)

c = 3e8;            % Speed of light (m/s)

m1 = 1; n1 = 0;     % TE10

m2 = 1; n2 = 1;     % TM11

k = 2*pi*f/c;

fc_TE10 = (c/2)*sqrt((m1/a)^2 + (n1/b)^2);

fc_TM11 = (c/2)*sqrt((m2/a)^2 + (n2/b)^2);

kc_TE10 = pi*sqrt((m1/a)^2 + (n1/b)^2);

kc_TM11 = pi*sqrt((m2/a)^2 + (n2/b)^2);

beta_TE10 = sqrt(complex(k^2-kc_TE10^2));
```

```
beta_TM11 = sqrt(complex(k^2-kc_TM11^2));
fprintf('TE10 vs TM11 MODE COMPARISON\n');
fprintf('-----------------------------------\n');
fprintf('TE10 cutoff frequency = %.4f GHz\n', ...
    fc_TE10/1e9);
fprintf('TE10 beta = %.4f + j(%.4f) rad/m\n', ...
    real(beta_TE10), imag(beta_TE10));
fprintf('\nTM11 cutoff frequency = %.4f GHz\n', ...
    fc_TM11/1e9);
fprintf('TM11 beta = %.4f + j(%.4f) rad/m\n', ...
    real(beta_TM11), imag(beta_TM11));
fprintf('\nOperating frequency = %.2f GHz\n', f/1e9);
if f > fc_TE10
    fprintf('TE10: PROPAGATING MODE\n');
else
    fprintf('TE10: BELOW CUTOFF\n');
end
if f > fc_TM11
    fprintf('TM11: PROPAGATING MODE\n');
else
    fprintf('TM11: BELOW CUTOFF (EVANESCENT)\n');
end
x = linspace(0,a,100);
```

```
y = linspace(0,b,60);

[X,Y] = meshgrid(x,y);

Ey_TE10 = sin(pi*X/a);

Hx_TE10 = Ey_TE10;

Ez_TM11 = sin(pi*X/a).*sin(pi*Y/b);

Ex_TM11 = -(pi/a)*cos(pi*X/a).*sin(pi*Y/b);

Ey_TM11 = -(pi/b)*sin(pi*X/a).*cos(pi*Y/b);

figure;

surf(X*1000,Y*1000,Ey_TE10,'EdgeColor','none');

xlabel('x (mm)');

ylabel('y (mm)');

zlabel('Normalized Ey');

title('TE10 Electric Field Distribution');

colorbar;

view(45,30);

figure;

surf(X*1000,Y*1000,Hx_TE10,'EdgeColor','none');

xlabel('x (mm)');

ylabel('y (mm)');

zlabel('Normalized Hx');

title('TE10 Magnetic Field Pattern (Hx)');

colorbar;

view(45,30);
```

```
figure;
surf(X*1000,Y*1000,Ez_TM11,'EdgeColor','none');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Ez');
title('TM11 Longitudinal Electric Field Pattern');
colorbar;
view(45,30);
figure;
subplot(1,2,1);
surf(X*1000,Y*1000,Ex_TM11,'EdgeColor','none');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Ex');
title('TM11 Ex Pattern');
colorbar;
view(45,30);
subplot(1,2,2);
surf(X*1000,Y*1000,Ey_TM11,'EdgeColor','none');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Ey');
title('TM11 Ey Pattern');
```

```
colorbar;

view(45,30);

Hx_TM11 = -(pi/b)*sin(pi*X/a).*cos(pi*Y/b);

Hy_TM11 = (pi/a)*cos(pi*X/a).*sin(pi*Y/b);

figure;

subplot(1,2,1);

surf(X*1000,Y*1000,Hx_TM11,'EdgeColor','none');

xlabel('x (mm)');

ylabel('y (mm)');

zlabel('Normalized Hx');

title('TM11 Hx Pattern');

colorbar;

view(45,30);

subplot(1,2,2);

surf(X*1000,Y*1000,Hy_TM11,'EdgeColor','none');

xlabel('x (mm)');

ylabel('y (mm)');

zlabel('Normalized Hy');

title('TM11 Hy Pattern');

colorbar;

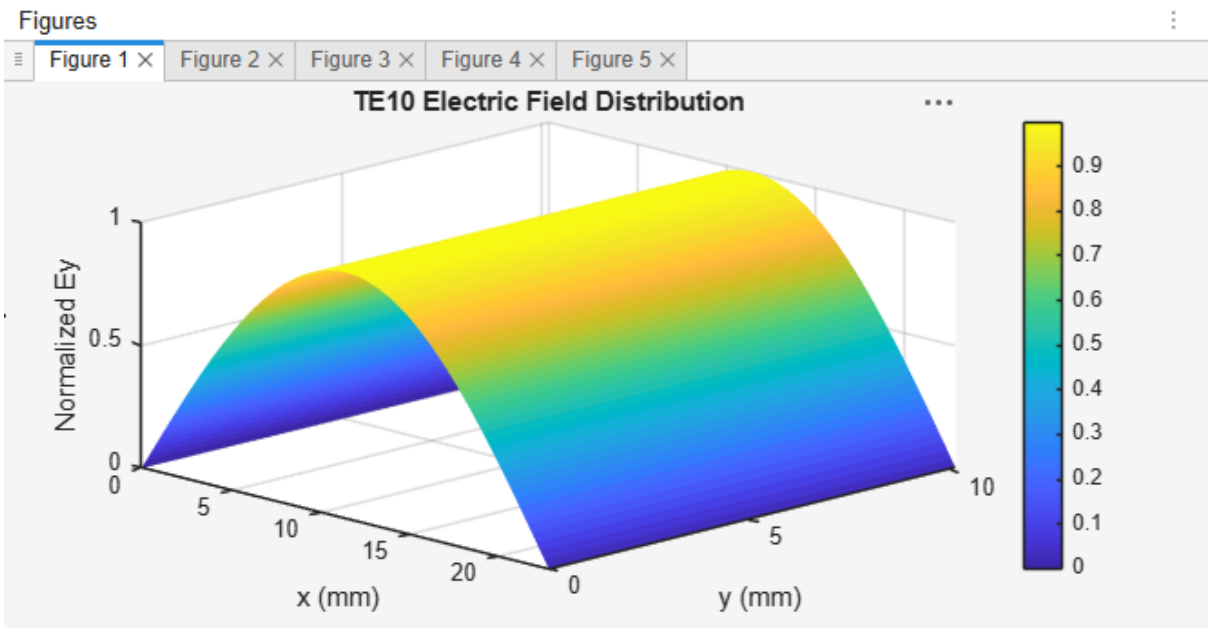
view(45,30);
```

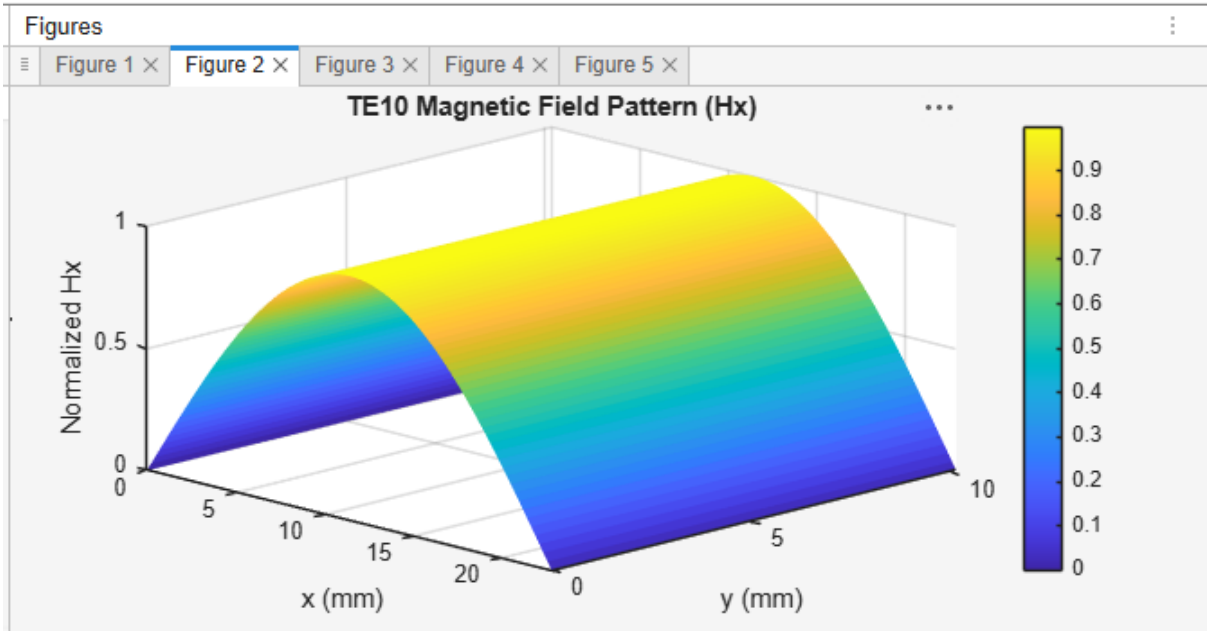
OUTPUT

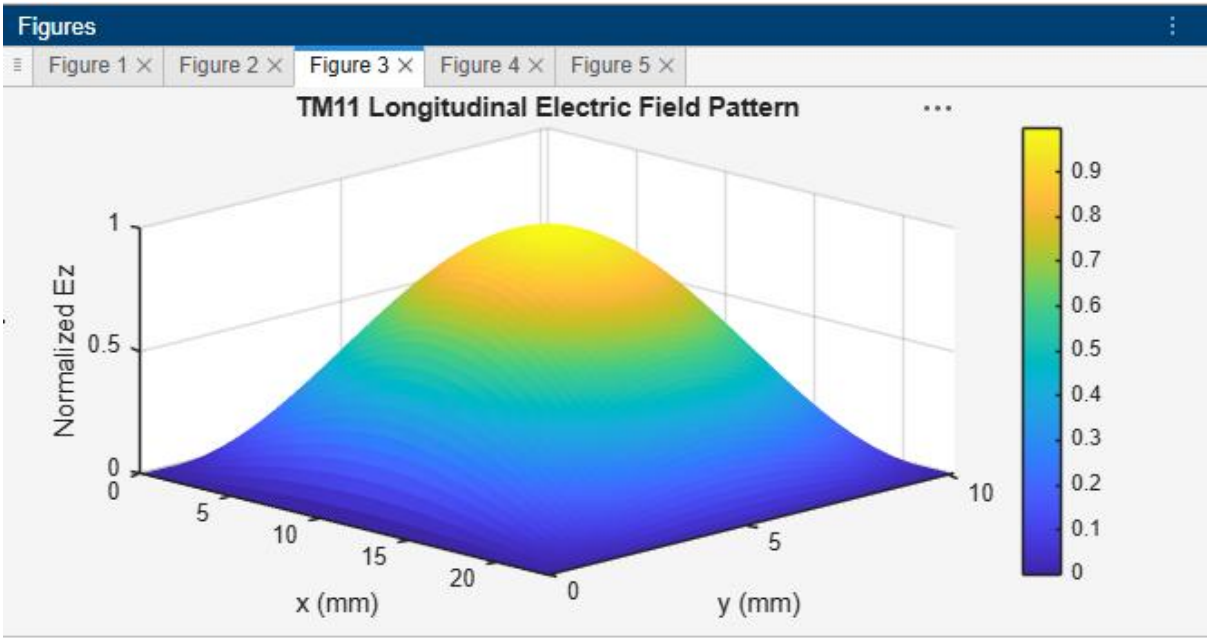
```
Command Window
TE10 vs TM11 MODE COMPARISON
------------------------------------
TE10 cutoff frequency = 6.5617 GHz
TE10 beta = 158.0462 + j(0.0000) rad/m

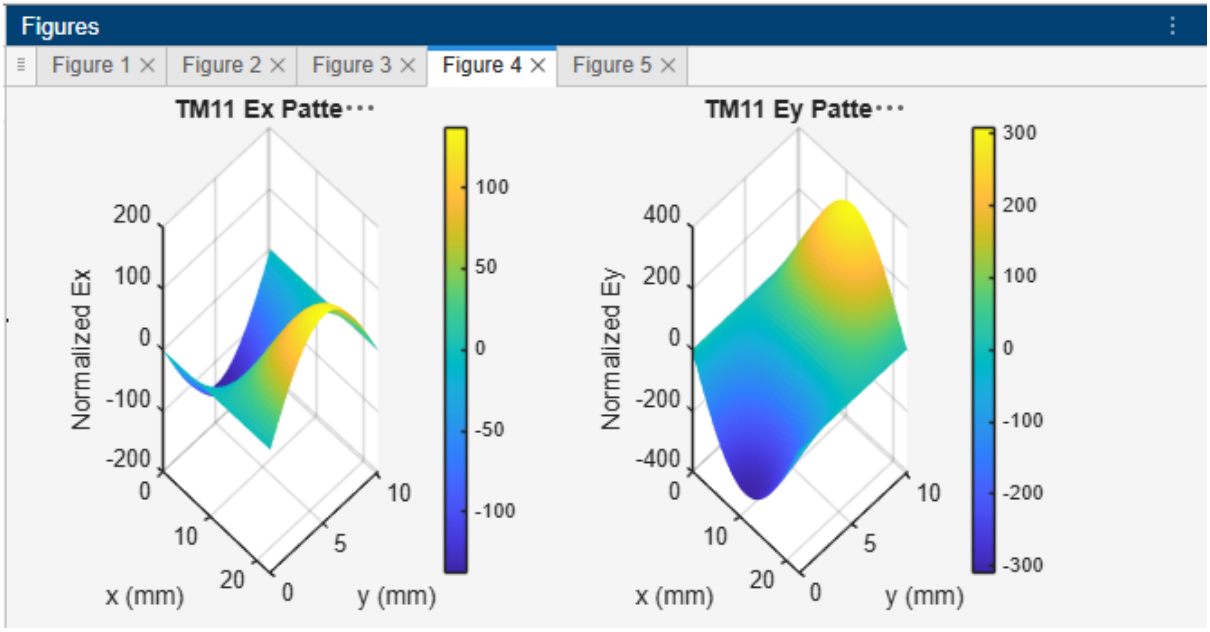
TM11 cutoff frequency = 16.1563 GHz
TM11 beta = 0.0000 + j(265.7694) rad/m

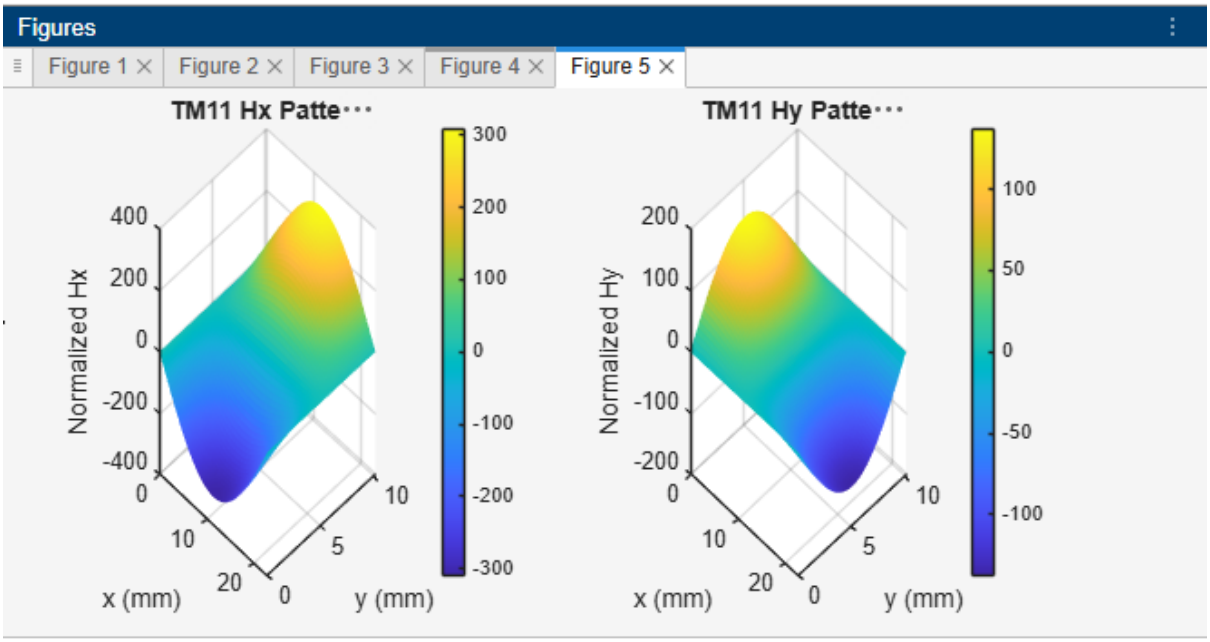
Operating frequency = 10.00 GHz
TE10: PROPAGATING MODE
TM11: BELOW CUTOFF (EVANESCENT)
>>
```




Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
TE10 Magnetic Field Pattern (Hx)
Normalized Hx
1
0.5
0
0
5
10
15
20
x (mm)
0
5
10
y (mm)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
TM11 Longitudinal Electric Field Pattern
Normalized Ez
1
0.5
0
0
5
10
15
20
x (mm)
0
5
10
y (mm)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0





RESULT

The TE_{10} and TM_{11} modes were analyzed for the WR-90 rectangular waveguide at 10 GHz. The theoretical cutoff frequencies were found to be approximately 6.562 GHz for TE_{10} and 16.15 GHz for TM_{11}. The TE_{10} mode propagates because its cutoff frequency is below the operating frequency, whereas the TM_{11} mode is below cutoff and exhibits evanescent behavior. The electric and magnetic field distributions were studied, confirming the different propagation characteristics of the two modes.